UME756 : Non-Traditional Machining Processes

L	T	P	Cr
2	0	2	3.0

Course Objective: To understand several non- traditional machining processes in micro and precision manufacturing field. To cultivate the ability to develop and implement new improved manufacturing processes resulting in creation and distribution of value in engineering applications. To impart knowledge about the significance of controlling process parameters for the optimal performance for newly developed engineering materials.

Syllabus

Introduction - Classification - process economy - Mechanical machining – Types - Ultrasonic machining (USM) - Abrasive Jet Machining (AJM) - Abrasive Flow Machining (AFM) - Water Jet Machining (WJM) - Operating principle - Process parameters – Design aspects, Applications - Limitations.

Laser materials processing - Laser types - Processes - Laser beam machining (LBM) – Laser cutting (LC) – Laser drilling (LD) - Laser marking and engraving (LM) – (Flip Learning component) Laser micromachining (LMM) - Laser engineered net shaping (LENS) - Applications – Limitations **Electro chemical machining** - Chemical material removal - Types - Electro chemical machining (ECM) - Electro chemical drilling (ECD) - Electro chemical grinding (ECG) - Electro chemical honing (ECH) - Shaped tube electrolytic machining - Operating principle - Process parameters - Applications - Limitations.

Thermo electrical machining - Types – Electrical discharge machining (EDM) - Electrical discharge wire cutting (EDWC) - Electron beam machining (EBM) - (Flip Learning component) Ion Beam Machining (IBM) -Plasma Arc Machining (PAM) - Operating principle - Process parameters

- Applications – Limitations

Laboratory Work

Experimental work pertaining to determination of effects of various process parameters on material removal rate, quality of machined surface, microstructure, optimization of process parameters in various Non- traditional machining processes.

Research assignment

Students will be divided in groups comprising of 4–5 students. Each group will be assigned with a separate research topic related to parametric analysis involved in various Non-traditional machining processes. Students will be required to go through the topics and recent developments from sources like reference books, journals etc. in the relevant field. Each group will be required to submit a report (and presentation) containing review of literature, summary, gaps in the existing literature, key findings etc.

Course Learning Objectives (CLO)

The students will be able to:

1. understand the contribution of non-traditional machining process in micro and precision manufacturing field.
2. summarize the merits and demerits of the non-traditional machining processes
3. analyze the processes and evaluate the role of each process parameter during machining of various advanced materials.
4. understand requirements to achieve maximum material removal rate and best quality of machined surface while machining various industrial engineering materials.

Reference Books

1. Abdel, H. and El-Hofy, G. “Advanced Machining Processes”, McGraw-Hill, 2005.
2. Jain, V.K., Advanced Machining processes, Allied Publishers Private Limited (2004).
3. Mishra, P.K., Non Conventional Machining, Narosa Publications (2006).
4. Groover, M.P. “Fundamentals of modern manufacturing processes - Materials, Processes and Systems”, 3rd Edition, John Wiley and Sons Inc., 2007.

Evaluation Scheme

S.No.	**Evaluation Elements**	**Weightage (%)**
1.	MST	25
2.	EST	45
3.	Sessionals (May include Research Assignments/Tutorials/Quizzes)	30